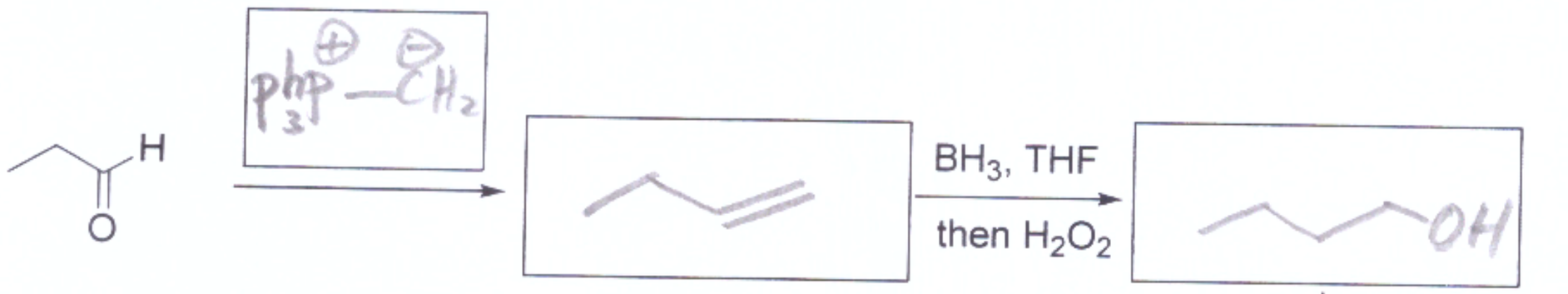
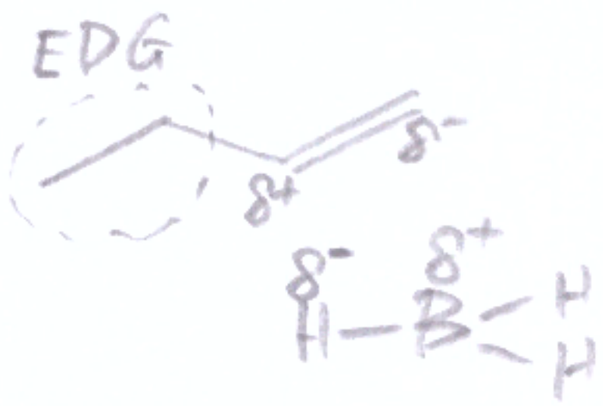
Name _____ Student number _____

Question 1

(1) Provide the structures of the reagents, intermediates and products in the boxes. (7 points)



(2) Explain the <u>regioselectivity</u> of the second step (the reaction using BH_3 in THF). (2 point)



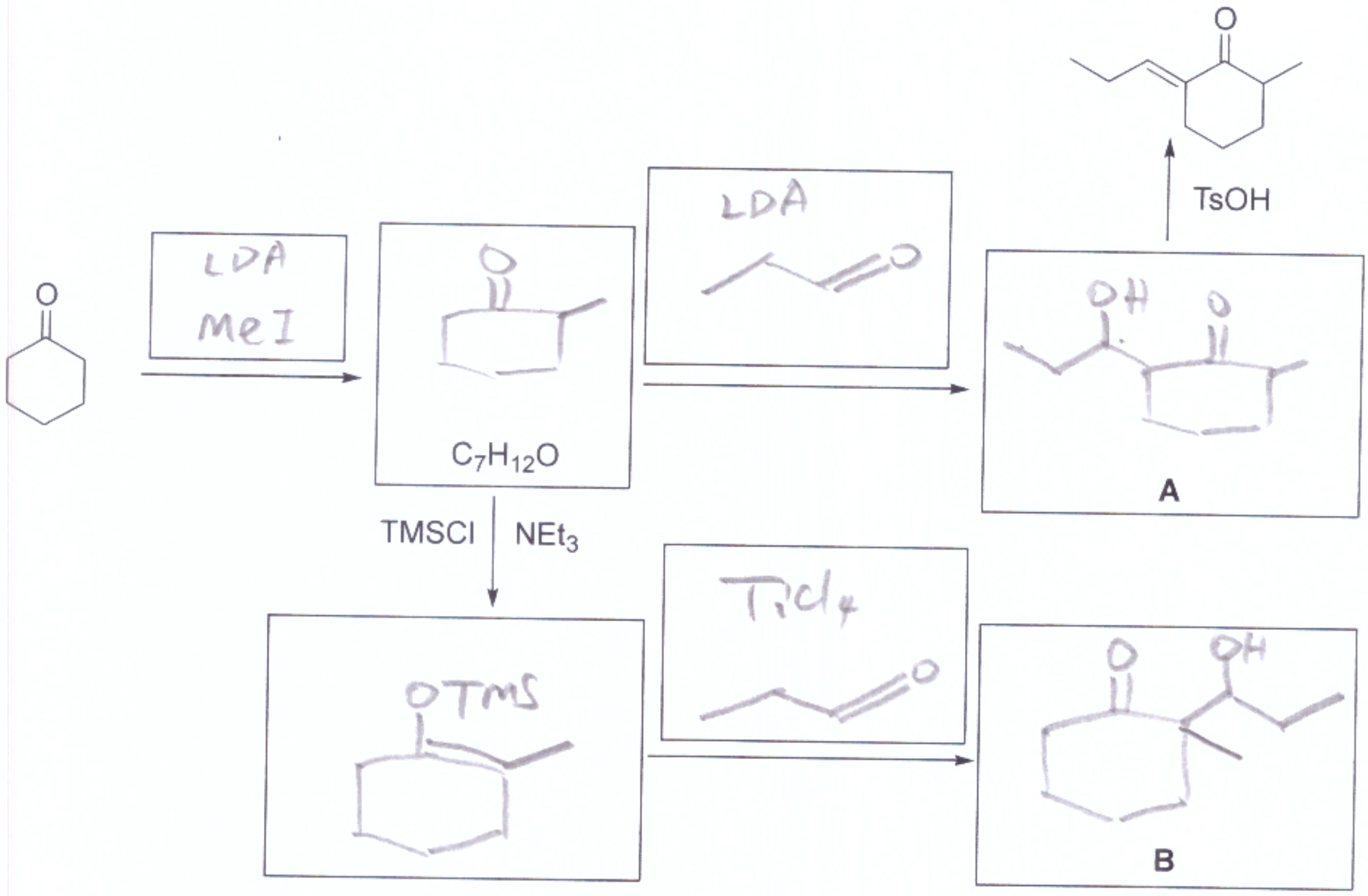
(3) Explain why $NaBH_3CN$ works better than $NaBH_4$ in the last step (the reaction with $MeNH_2$). (1 point)

$NaBH_4$ reduces ketone directly.

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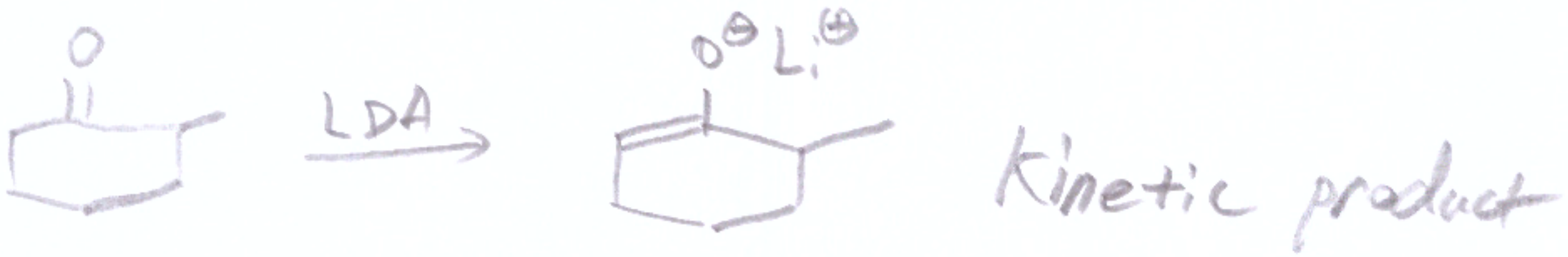
Question 2

(1) Provide the structures of the reagents, intermediates and products in the boxes. (7 points)



A and **B** are structural (or constitutional) isomers

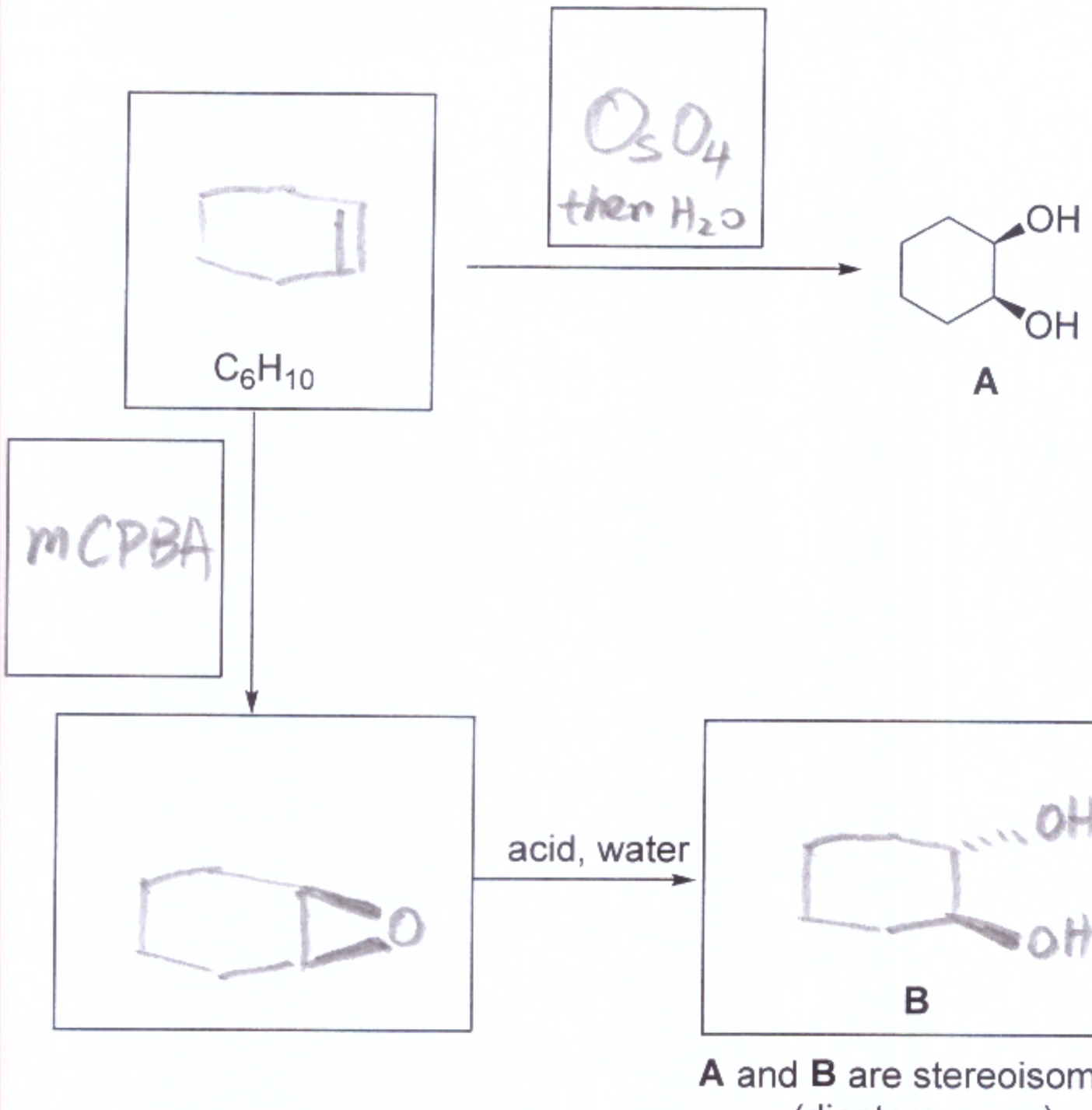
(2) Explain the <u>regioselectivity</u> for the formation of product **A** and product **B** under the respective reaction conditions that you have provided. (2 points)



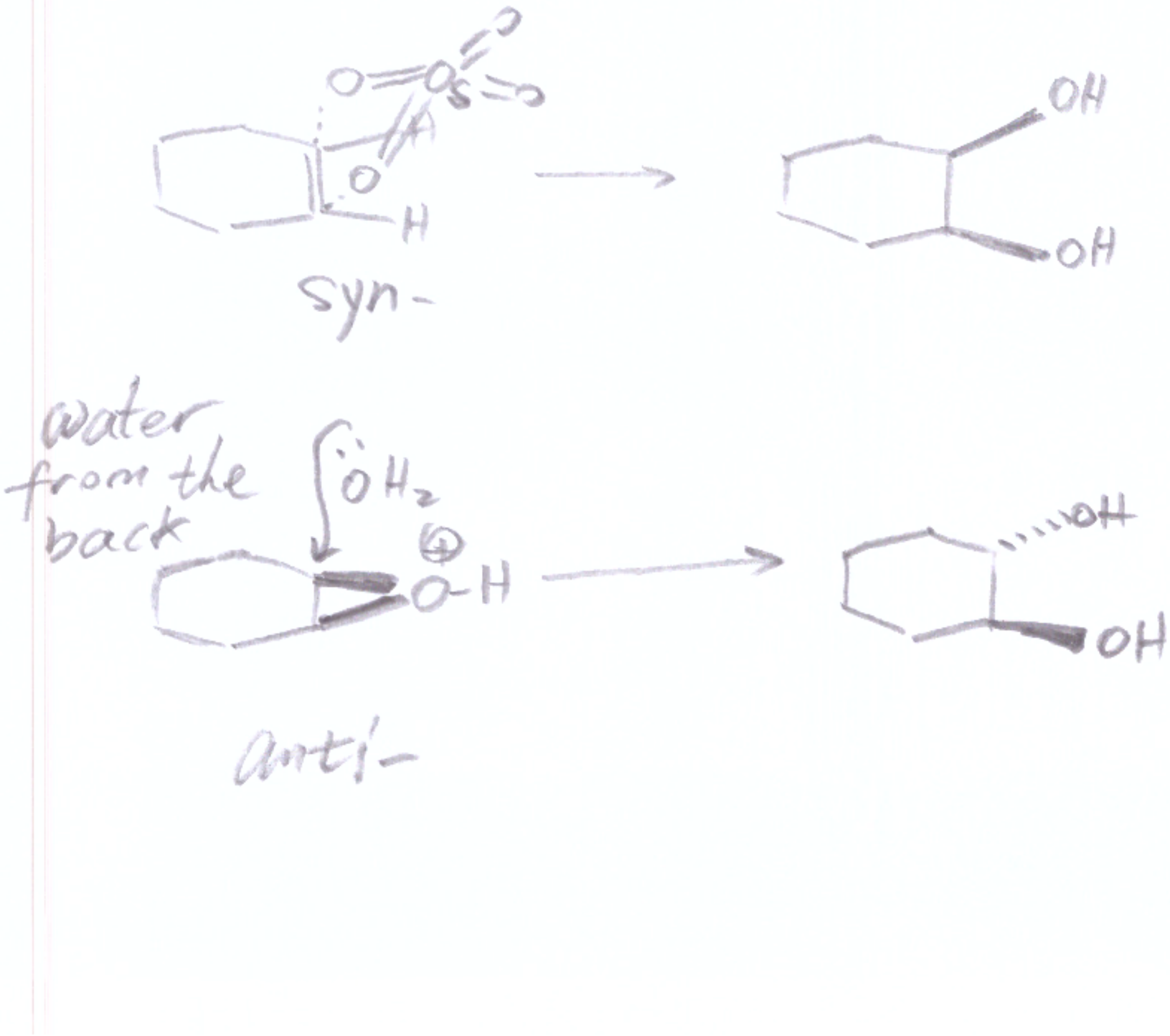
Name _____ Student number_____

Question 3

(1) Provide the structures of the reagents, intermediates and products in the boxes. (5 points)



(2) Explain the <u>stereoselectivity</u> for the formation of product **A** and product **B** under the respective reaction conditions that you have provided. (2 points)

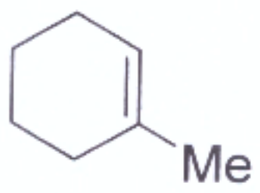
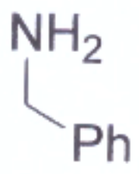
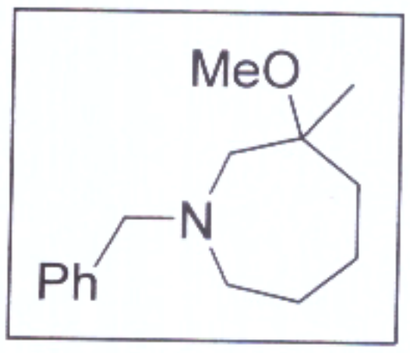


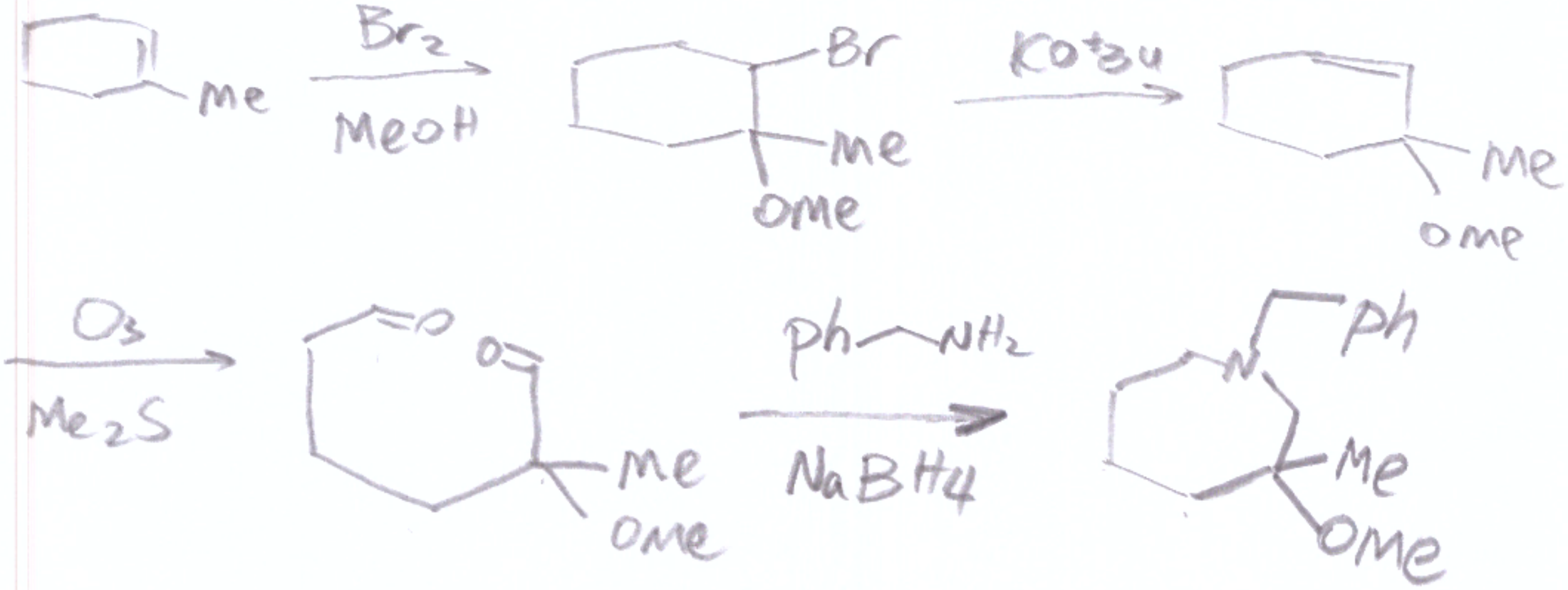
Name _____ Student number_____

Question 4

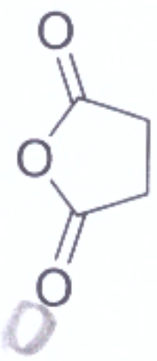
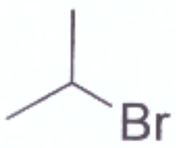
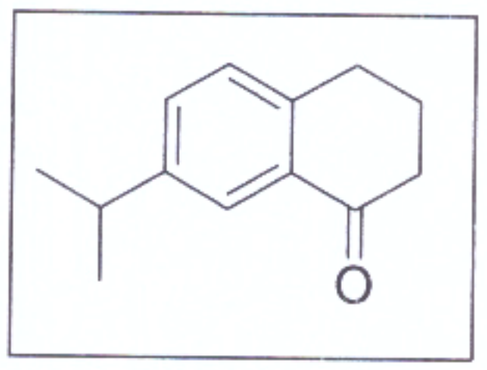
Provide efficient synthetic routes for the following target molecules in the boxes using the starting materials specified. You may also use readily available reagents or reactants as we have discussed in the lectures in addition to these starting materials. Clearly show the products and reaction conditions of each step.

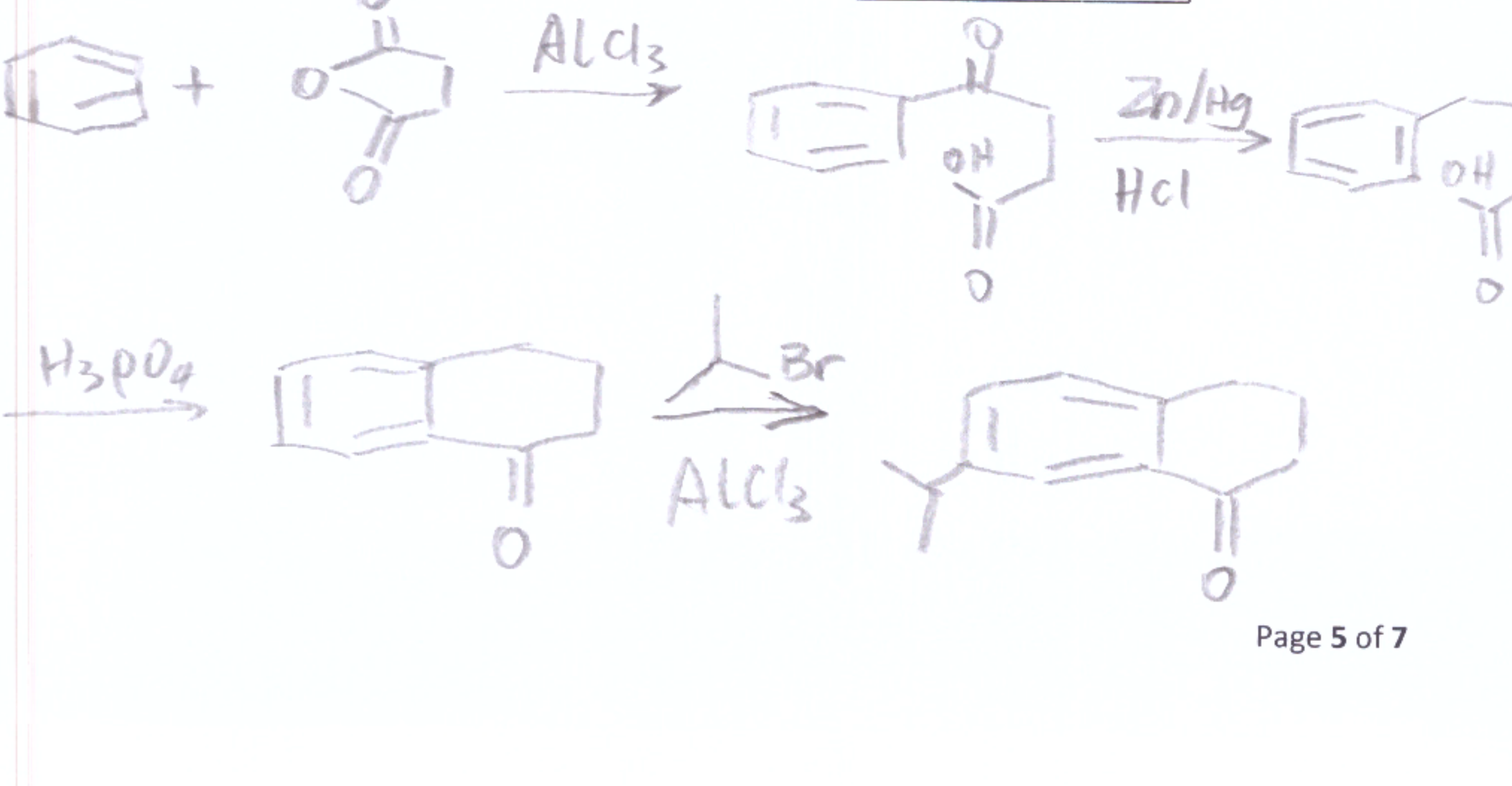
(1) (5 points)

from  and  to 



(2) (5 points)

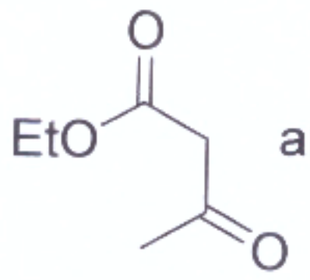
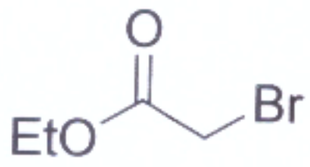
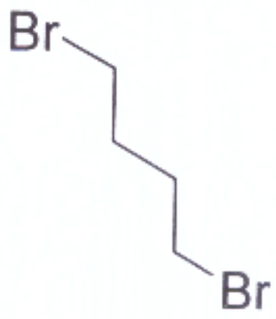
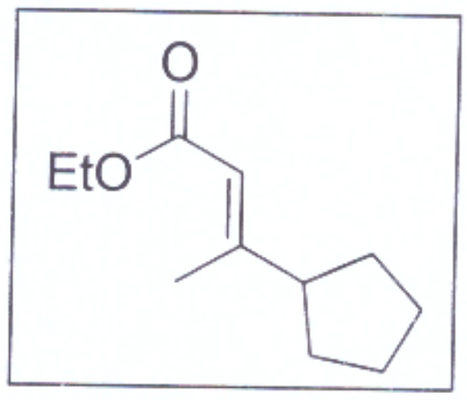
from  and  and  to 

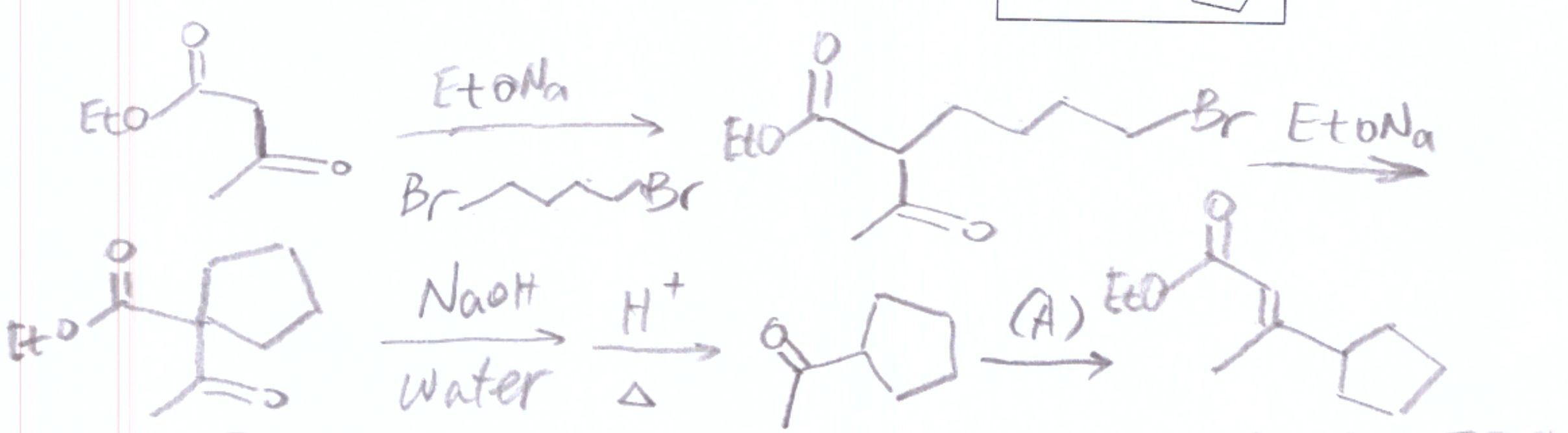


Name _____ Student number _____

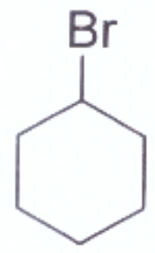
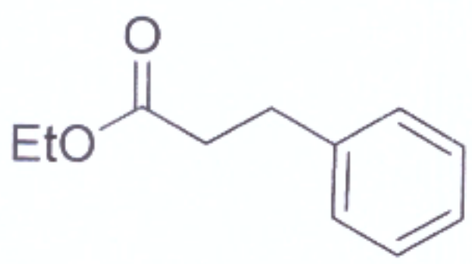
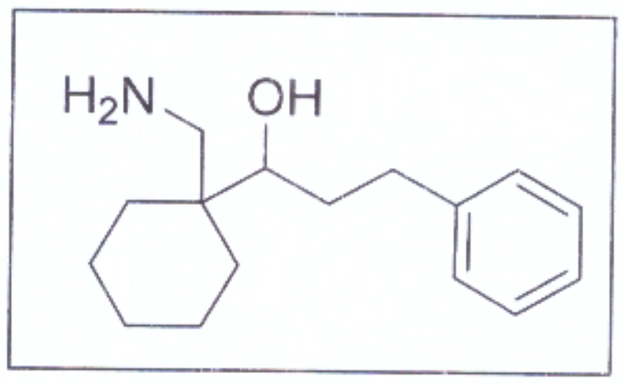
Question 4 (continued)

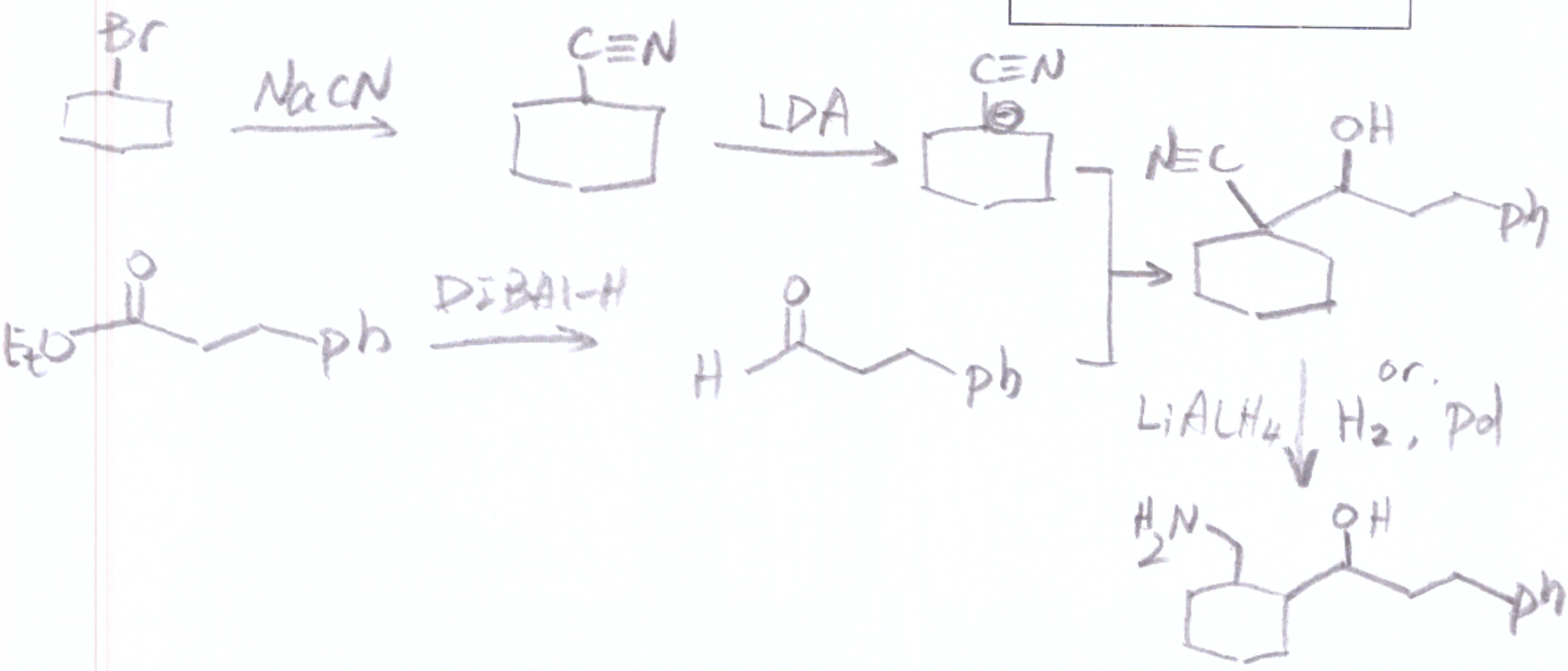
(3) (5 points)

from  and  and  to 



(4) (5 points)

from  and  to 

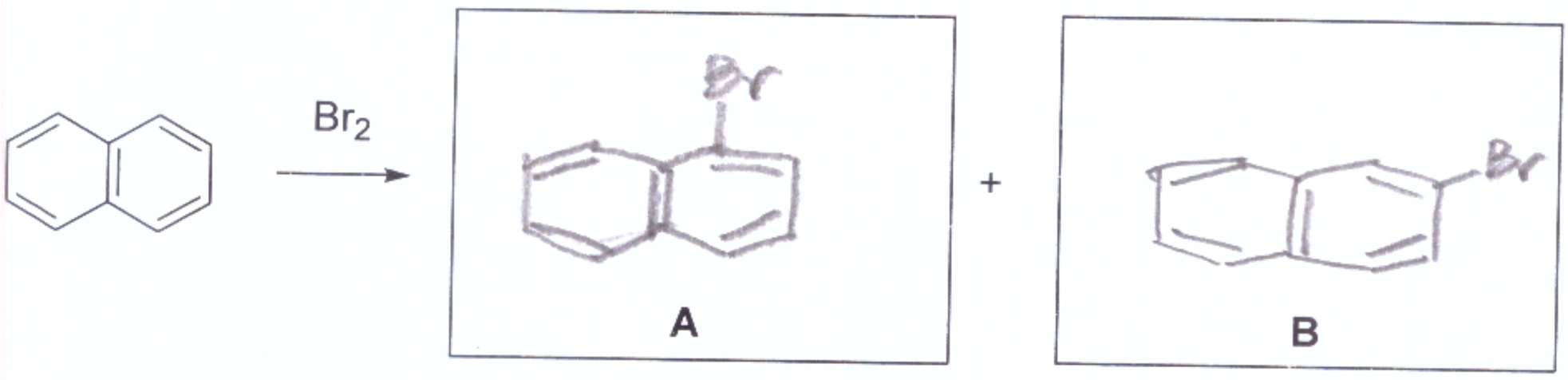


Name _____ Student number _____

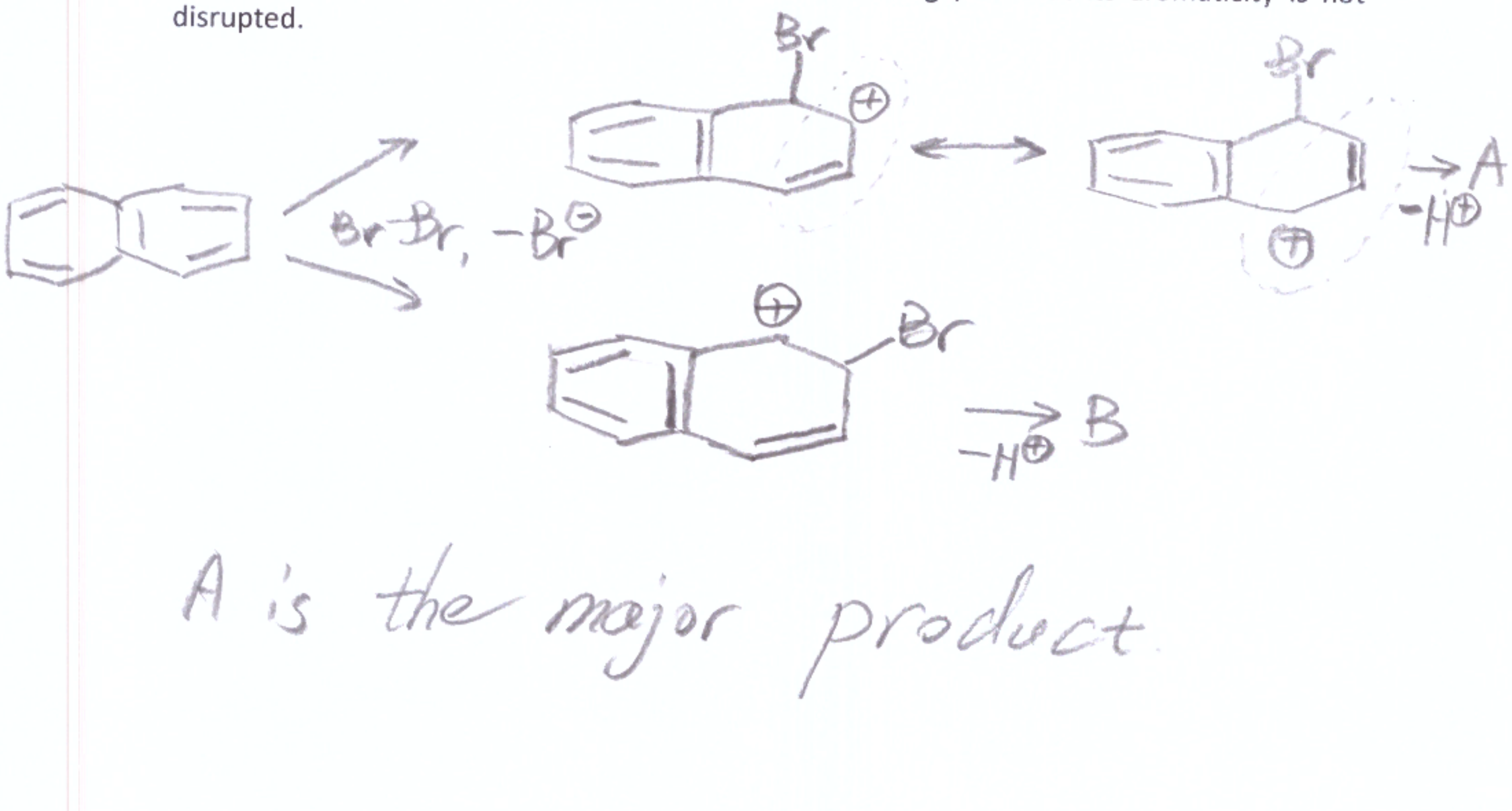
Question 5

Mono-bromination of naphthalene gives two products **A** and **B** as regioisomers.

(a) Provide the chemical structures of the two mono-bromination products. (2 points)



(b) Draw the structures of arenium ions leading to the formation of the *mono*-bromination products. Based on the possible resonance structures of the corresponding arenium ions, which regioisomer is the major *mono*-bromination product? (2 points)
Hint: when drawing the resonance structures, you may assume the "1,3,5-cyclohexatriene" structure of the remaining benzene unit does not change, so that its aromaticity is not disrupted.



A is the major product